

KATHMANDU UNIVERSITY
University Examination
February/March 2022

Level : BDS

Subject: Microbiology

Exam : BDS I

Paper : 1

Time : 2 hour 30 minutes

Marks : 50

Date : March 10, 2022

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book

Write brief, relevant and legible answers

Illustrate your answer with well labeled diagrams whenever required

All questions are compulsory

SECTION "B" [25 marks]

List four immunoglobulins. Describe the structure and functions of IgG. [1+2+3=6]

Enumerate the differences between Gram positive and Gram negative bacteria with examples. [5]

Write in brief about the life cycle of *Leishmania donovani*. How will you perform the laboratory diagnosis of infection of blood parasites? [2+3=5]

Write short notes on: [3×3=9]

- Morphological classification of fungi
- Epstein Barr Virus
- Tabulate the differences between Exotoxin and Endotoxin

SECTION "C" [25 marks]

Enumerate the pathogens that cause pneumonia. Write in brief about pathogenesis and laboratory diagnosis of pulmonary aspergillosis. [2+2+2=6]

What are the causative agents of different types of infective endocarditis? Write down its pathogenesis. [2+3=5]

Discuss about the pathogenesis and laboratory diagnosis of Streptococcal sore throat. [2+3=5]

Write short notes on: [3×3=9]

- Pathogenesis of *Mycobacterium tuberculosis*
- Antigenic variation in influenza virus
- Laboratory diagnosis of Rheumatic Heart disease

KATHMANDU UNIVERSITY
August/ September 2021

Subject: Microbiology

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : September 1, 2021

Paper : I

Marks : 50

Level : BDS

Exam : BDS I

Time : 2 hr. 30

Date : March

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

- Answer each
- Write brief
- Illustrate
- All questions

SECTION "B"
[25 marks]

1. Draw well labelled diagram of generalized bacterial cell. [2+3=
2. Describe the laboratory diagnosis of malaria.
3. Write down the antigen and antibody test used for diagnosis of infectious diseases. Explain ELISA test. [2+3=
4. Write short notes on:
a. Antimicrobial susceptibility test. [3×3=
- b. Contribution of Louis Pasteur.
- c. Life cycle of *Leishmania donovani*

1. Define
2. Describe
3. Describe
4. Write
a. B
b. T
c. M

SECTION "C"
[25 marks]

5. Describe the classification methods of *streptococci* write pathogenesis rheumatic heart disease. [3+3=
6. Write virulence factors of *Streptococcus pneumoniae* and its laboratory diagnosis. [2+3=
7. Describe the pathogenesis and laboratory diagnosis of *Bordetella pertussis*. [3+2=
8. Write short notes on:
a. Diphtheria. [3×3=
- b. Infective Endocarditis.
- c. Laboratory diagnosis of pulmonary tuberculosis.

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KATHMANDU UNIVERSITY
March/April 2021

Level : BDS

Subject: Microbiology

Exam : BDS I

Paper : I

Time : 2 hr. 30 minutes

Marks : 50

Date : March 31, 2021

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Define sterilization. Explain moist heat methods of sterilization. [1+5=6]
2. Describe different types of immunity with examples. [5]
3. Describe the laboratory diagnosis of blood bone parasites. [5]
4. Write short notes on: [3×3=9]
 - a. Bacterial growth curve.
 - b. Tabulate the differences between *P. falciparum* and *P. vivax*.
 - c. Methods of anaerobiasis.

SECTION "C"
[25 marks]

5. Describe the pathogenesis and diagnosis of pulmonary tuberculosis. [6]
6. Explain laboratory diagnosis of throat infection. [5]
7. Describe the pathogenesis and laboratory diagnosis of infective endocarditis. [5]
8. Write short notes on: [3×3=9]
 - a. Pathogenesis of rheumatic heart disease.
 - b. Pathogenesis of *Streptococcus pneumoniae*.
 - c. Draw well labelled diagram of influenza virus.

Subject: Microbiology

Level : BDS

Paper : I

Exam : BDS I (new course)

Marks : 50

Time : 2 hour 30 minutes

Date : January 9, 2019

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Define antibiotics? Explain mechanism of action of antimicrobial agents. [1+5=6]
2. Differentiate between antiseptics and disinfection. Mention types of chemical agents with their use. [2+3=5]
3. What are immunoglobulins? Explain the classes of immunoglobulins. [1+4=5]
4. Write short notes on:
a. Type 4 hypersensitivity [3×3=9]
b. Kalaazar
c. Bacteraemia

SECTION "C" [25 marks]

5. Describe pathogenesis and laboratory diagnosis of *M. tuberculosis*. [2+3=6]
6. List the normal flora of upper respiratory tract. Describe pathogenesis and laboratory diagnosis of diphtheria. [1+2+2=5]
7. Describe pathogenesis and lab diagnosis of rheumatic heart disease. (80 re throat) [3+2=5]
8. Write short notes on:
a. Lab diagnosis of infective endocarditis [3×3=9]
b. Lab diagnosis of pulmonary aspergillosis
c. *Streptococcus pneumoniae*

UNIVERSITY
July/August 2019

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : August 8, 2019

Subject: Microbiology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. List four immunoglobulins. Describe the structure and functions of IgM. [2+2+2=6]
2. What is mode of infection of *kala-azar* and describe lab diagnosis of *kala-azar*. [1+4=5]
3. Define hypersensitivity reaction? Describe Type-I hypersensitivity reaction. [1+4=5]
4. Write short notes on: [3×3=9]
 - a. Autoclave
 - b. Hear air oven
 - c. Disinfection

SECTION "C" [25 marks]

5. Draw a diagram of influenza virus. Explain antigenic drift and antigenic shift? [2+2+2=6]
6. Describe the pathogenesis and lab diagnosis of rheumatic fever. [3+2=5]
7. Describe the laboratory diagnosis of tuberculosis. [5]
8. Write short notes on: [3×3=9]
 - a. Plasmodium falciparum
 - b. Toxoplasma infection
 - c. Bacterial endocarditis

KATHMANDU UNIVERSITY
July/August 2018

Level : BDS

Exam : BDS I (new course)

Time : 2 hour 30 minutes

Date : July 13, 2018

Subject: Microbiology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B"
[25 marks]

1. Draw a bacterial growth curve and describe each phase. [2+4=6]
2. Discuss in brief the laboratory diagnosis of malaria. [5]
3. Discuss in brief the transmission cycle of *Leishmania donovani*. [5]
4. Write short notes on:
a. Conjugation
b. Prevention of Nosocomial infections
c. Hypersensitivity [3×3=9]

SECTION "C"
[25 marks]

5. Discuss the laboratory diagnosis of community acquired pneumonia. [6]
6. Write the mechanism of action of Diphtheria Toxin. [5]
7. Discuss the pathogenesis of Rheumatic heart disease. [5]
8. a. Define and list the causative agents of Infective Endocarditis. [1+2=3]
b. Discuss the 'sample collection' for diagnosis of pulmonary tuberculosis. [3]
c. List the virulence factor of *Bordetella pertussis*. [3]

KATHMANDU UNIVERSITY
December 2016/January 2017

Subject: BDS

Course: BDS I (new course)
Duration: 2 hour 30 minutes

Date: January 6, 2017

Subject: Microbiology

Paper : 1

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Write brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "B" [25 marks]

Describe the contribution of Robert Koch and what are the Robert Koch postulates?

[3+3=6]

Define an antigen. Describe the conditions of the antigenicity.

[1+4=5]

Describe the life cycle and laboratory diagnosis of *Plasmodium falciparum*.

[3+2=5]

Write short notes on:

[3×3=9]

- Septicaemia
- What are the different modes of action of antibiotics?
- Laboratory diagnosis of Epstein Barr virus

SECTION "C" [25 marks]

List the pathogens of respiratory tract infection. Describe the laboratory diagnosis of *Streptococcus pyogenes*.

[2+4=6]

List etiologic agents of endocarditis. Describe the laboratory diagnosis of infective endocarditis.

[2+3=5]

Describe pathogenesis and laboratory diagnosis of *Streptococcus pneumoniae*.

[3+2=5]

Write short notes on:

[3×3=9]

- Diagnosis of Influenza virus infection
- Bordetella pertussis*
- Pathogenesis of rheumatic heart disease

KATHMANDU UNIVERSITY
May/June 2016

Level : BDS

Exam : BDS I (new course)

Time : 2 hr. 30 minutes

Date : May 29, 2016

Subject: Microbiology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "B" [25 marks]

1. Define sterilization. Describe the principle and application of an autoclave. [1+3+2=6]
2. Define an antibody. Describe the types of antibodies. [1+4=5]
3. Define vaccine. Describe the types of vaccines used in routine childhood vaccination. [1+4=5]
4. Write short notes on:
 - a. Antibiotic sensitivity test
 - b. Replication of virus
 - c. Anaerobiosis[3×3=9]

SECTION "C" [25 marks]

5. List the organism causing lower respiratory tract infection. Describe pathogenesis and laboratory diagnosis of *Streptococcus pneumoniae*. [2+2+2=6]
6. What is a causative agent rheumatic heart disease? Write its pathogenesis and laboratory diagnosis. [1+2+2=5]
7. Describe the pathogenesis and laboratory diagnosis of streptococcal sore throat. [3+2=5]
8. Write short notes on:
 - a. Laboratory diagnosis of pulmonary tuberculosis
 - b. Morphology of influenza virus
 - c. *Histoplasma capsulatum*[3×3=9]

BDS
BDS I (new course)
2 hour 30 minutes
October 24, 2016

Subject: Microbiology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book
Write brief, relevant and legible answers
Illustrate your answer with well labeled diagrams whenever required
All questions are compulsory

SECTION "B" [25 marks]

List the different methods of sterilization. Describe the principle and use of Hot air oven. [2+4=6]

Describe the life cycle and laboratory diagnosis of *Wuchereria bancrofti*. [3+2=5]

Define antibody. List various types of antibodies? Draw a labeled diagram of IgM. [1+1+3=5]

Write short notes on: [3×3=9]

- Distinguish between nomenclature and classification
- Anaerobiosis
- Laboratory diagnosis of Epstein Barr virus

SECTION "C" [25 marks]

1. List the organisms causing upper respiratory tract infection. Describe the pathogenesis and laboratory diagnosis of pulmonary tuberculosis. [1+5=6]

2. List etiologic agents of endocarditis. Describe the laboratory diagnosis of infective endocarditis. [2+3=5]

3. Describe the pathogenesis and laboratory diagnosis of streptococcal sore throat. [3+2=5]

4. Write short notes on: [3×3=9]

- Diagnosis of Influenza virus infection
- Laboratory diagnosis of *Cryptococcus neoformans*
- Pathogenesis of rheumatic heart disease

KATHMANDU UNIVERSITY
2015

Level : BDS
Exam : BDS I (new course)
Time : 2 hr. 30 minutes
Date : September 16, 2015

Subject: Microbiology

Paper : I

Marks : 50

INSTRUCTIONS TO THE CANDIDATE

Answer each section in a separate answer book

Write brief, relevant and legible answers

Illustrate your answer with well labeled diagrams whenever required

All questions are compulsory

SECTION "B" [25 marks]

Draw labeled diagrams depicting the differences in the cell wall structure of Gram positive and Gram negative bacteria.

[5]

List the different methods of transfer of genetic material in bacteria. Discuss in brief any one of these.

[2+4=6]

Discuss the life cycle of Plasmodium species.

[5]

Write short notes on:

- Cryptococcus neoformans
- Laboratory diagnosis of filariasis
- Properties of oncogenic viruses

[3×3=9]

SECTION "C" [25 marks]

List four bacteria causing community acquired pneumonia. Discuss in brief the laboratory diagnosis of pneumonia.

[2+4=6]

Briefly discuss the pathogenesis of pharyngeal diphtheria.

[5]

Define infective endocarditis (IE). Discuss the laboratory diagnosis of IE.

[1+4=5]

Write short notes on:

[3×3=9]

- Pathogenesis of Acute Rheumatic Fever
- How pandemics of influenza occur
- Cultural characteristics of Mycobacterium tuberculosis.

KATHMANDU UNIVERSITY
2015

Level : BDS

Exam : BDS I (old course)

Time : 3 hrs.

Date : September 16, 2015

Subject: Microbiology

Paper : I

Marks : 70

INSTRUCTIONS TO THE CANDIDATE

- Answer each section in a separate answer book
- Write brief, relevant and legible answers
- Illustrate your answer with well labeled diagrams whenever required
- All questions are compulsory

SECTION "A" [35 marks]

1. Describe the terms sterilization, disinfection and antiseptic. Explain the principle and operation of an autoclave. [3+2+3=8]
2. Explain the composition and functions of bacterial cell wall and explain the bacterial growth curve with diagram. [2+2+3=7]
3. Define antibiotics and explain the types of antibiotics used in treatment of human infections. [5+5=10]
4. Distinguish between antigen and antibodies. List the conditions of antigenicity. Explain the types and roles of different vaccines in protections against infections. [2+3+5=10]

SECTION "B" [35 marks]

5. List pathogens of the lower respiratory tract. Describe pathogenesis and laboratory diagnosis of bacterial pneumonia. [2+4+4=10]
6. List the possible etiological agents for upper respiratory infection. Explain the pathogenesis and laboratory diagnosis of seasonal influenza virus. [2+4+4=10]
7. List etiological agents and risk factors and laboratory diagnosis of infective endocarditis. [2+2+4=8]
8. Describe the pathogenesis and laboratory diagnosis of pulmonary tuberculosis. [3+4=7]